EASWARI ENGINEERING COLLEGE
(Autonomous)
Bharathi Salai, Ramapuram, Chennai – 89

DIGITAL HEALTHCARE PLATFORM

Integrated Appointment, Diagnostic Testing & Healthcare Journey Management

Department of INFORMATION TECHNOLOGY

PRESENTED BY

MADHIARASAN M

Under the guidance of

Grandhi Sai Lakshmi

CONTENTS

- Abstract
- Introduction
- Literature Survey
- Existing System and Disadvantages
- Proposed System and Advantages
- System Requirements
- System Architecture
- References

ABSTRACT

The proposed system is a digital healthcare platform that allows patients to manage major healthcare activities through a single online application. Patients can book diagnostic tests, schedule appointments, make payments, receive reports digitally, and track their healthcare journey from booking to result delivery. The platform aims to reduce manual processes, waiting time and fragmented service management while improving accessibility and convenience. The design follows the concept of an integrated digital health platform that connects healthcare applications and information services.

INTRODUCTION

Healthcare services are often distributed across separate appointment, diagnostic, payment and reporting processes. Patients may need repeated visits, physical paperwork and multiple communication channels to complete a single healthcare journey.

The proposed platform brings these services together in one digital system. It provides patient registration, appointment scheduling, diagnostic test booking, payment processing, report delivery and journey tracking. A centralized backend and database maintain the required information and allow authorized healthcare staff to manage services efficiently.

LITERATURE SURVEY

AUTHOR	YEAR OF PUBLICATION	DESCRIPTION	PROS	CONS
WHO & ITU	2020	Presented a digital health platform architecture for building interoperable and integrated national digital health systems	Supports interoperability, integration, scalability, and reuse of digital health services	Primarily focuses on infrastructure and architecture rather than a complete patient-facing system
WHO	2023	Developed a classification framework for digital health interventions, services, and applications, covering different healthcare and digital-service use cases	Provides standardized terminology and helps organize and describe digital health applications consistently	It is mainly a classification framework and does not provide a complete implementation of a healthcare management system
WHO SMART Base	2021	Proposed a global strategy for digital health focusing on the development, implementation, and scaling of digital technologies in healthcare systems	Provides strategic guidance for interoperability, digital transformation, and sustainable healthcare technology adoption	Provides high-level strategic guidance rather than detailed software implementation or a complete application

EXISTING SYSTEM

- Appointment booking and diagnostic test services are often handled separately.
- Patients may need physical visits for registration, booking, payment or report collection.
- Different healthcare providers may maintain separate patient records.
- Communication about appointment status and diagnostic reports may depend on manual processes.
- Limited integration makes it difficult for patients to track the complete healthcare journey.

DISADVANTAGES

- Time-consuming manual procedures
- Long waiting and repeated hospital visits
- Multiple systems or channels for different healthcare services
- Difficulty tracking appointments, tests and reports
- Paper-based or delayed report delivery
- Data duplication and possible manual-entry errors
- Limited visibility of the patient's complete healthcare journey

PROPOSED SYSTEM

- Patient registration and secure login
- Online doctor appointment scheduling
- Diagnostic test booking with available slots
- Online payment and transaction status
- Digital diagnostic report delivery
- Patient dashboard for appointments, tests and reports
- Healthcare journey tracking from booking to result delivery
- Admin dashboard for managing patients, appointments, tests and reports
- Centralized database for authorized healthcare information

ADVANTAGES

- Centralized healthcare services in one platform
- Reduced waiting time and unnecessary visits
- Faster access to diagnostic reports
- Convenient online payment and booking
- Improved visibility of appointment and test status
- Reduced paperwork and administrative effort
- Better coordination between patients, providers and diagnostic services
- Scalable architecture for adding future healthcare services

SYSTEM REQUIREMENTS

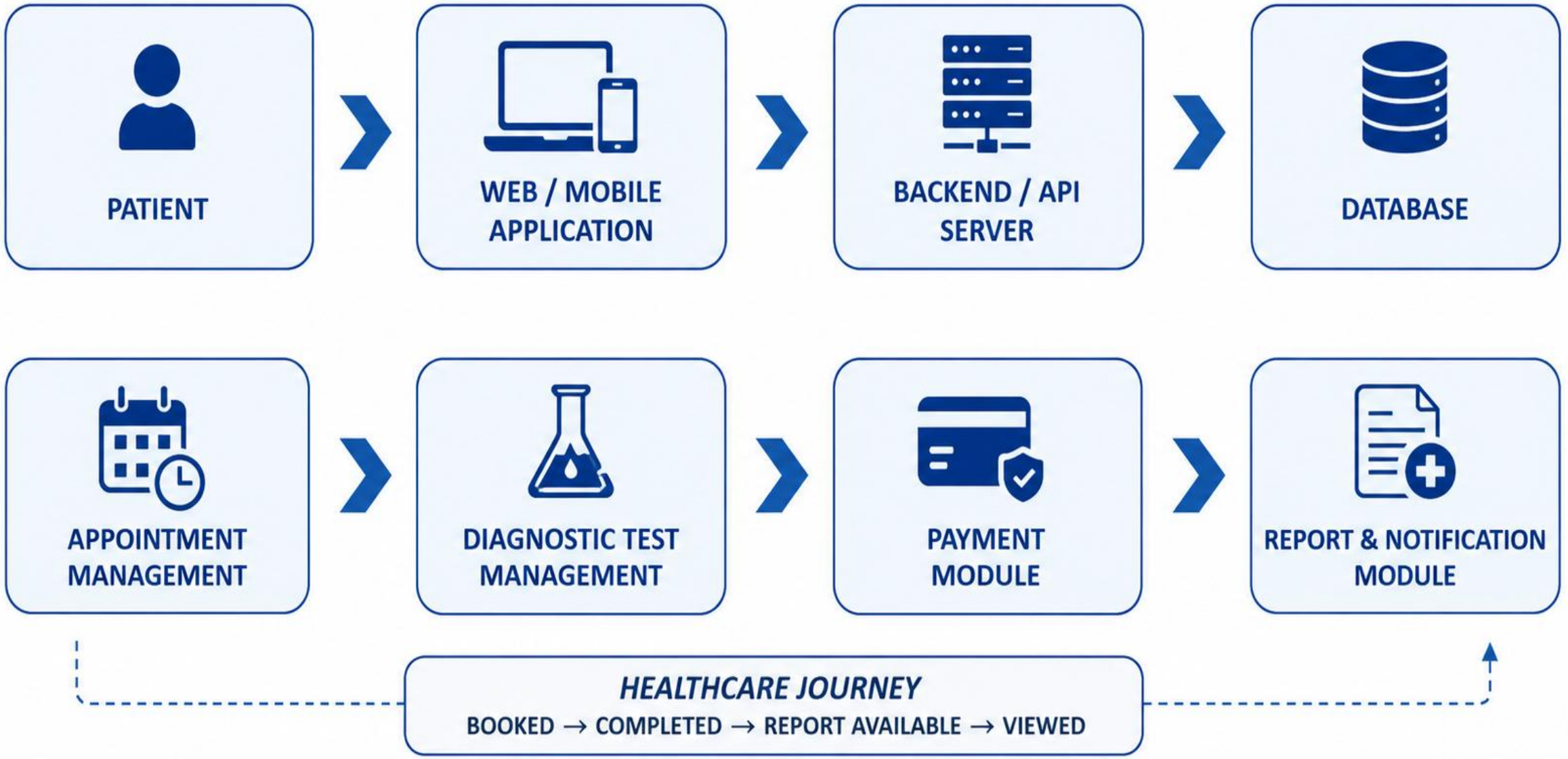
HARDWARE SPECIFICATIONS:

PROCESSOR : Intel Core i3 or above
RAM : Minimum 4 GB
HARD DISK : 20 GB or above
INPUT DEVICE : Standard Keyboard and Mouse
MONITOR : Minimum Resolution 1920×1080

SOFTWARE SPECIFICATIONS:

OPERATING SYSTEM : Windows / Linux
FRONTEND : React.js, HTML, CSS, JavaScript
BACKEND : Node.js, Express.js
DATABASE : MongoDB
BROWSER : Chrome, Edge or Firefox
INTERNET CONNECTION : Required

SYSTEM ARCHITECTURE



PATIENT HEALTHCARE JOURNEY

- 1. Patient registers / logs in
- 2. Selects doctor appointment or diagnostic test
- 3. Chooses available date and time slot
- 4. Confirms booking and completes payment
- 5. Receives appointment / test status updates
- 6. Diagnostic result is processed and uploaded
- 7. Patient receives and views the report online
- 8. Healthcare journey and previous records remain available in the dashboard

REFERENCES

[1] World Health Organization & International Telecommunication Union, Digital Health Platform Handbook: Building a Digital Information Infrastructure (Infostructure) for Health, 2020.

[2] World Health Organization, Classification of Digital Interventions, Services and Applications in Health, 2nd ed., 2023.

[3] World Health Organization, SMART Base – Electronic Medical Record System and Laboratory Information System resources, 2026.

[4] World Health Organization, Digital Health – Global Strategy and digital health resources.

THANK YOU